

Ila Rehman

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Summary

MCA graduate and self-taught AI engineer with a physics background, focused on shipping LLM-powered applications. Built a multi-tenant AI sales agent (**LangGraph**, **FastAPI**) and a production **RAG** API (**FastAPI**, **LangChain**, **FAISS**, **GPT-4o**), and currently build event-driven **AWS** microservices and **LLM**-powered agents in production as a software engineer intern.

Experience

Manrke

April 2026 – Present

Software Engineer Intern

Remote

- Design and build distributed, event-driven microservices on **AWS** (**Lambda**, **ECS**, **Step Functions**, **SQS/SNS**, **S3**, **DynamoDB**, **RDS**) for an end-to-end document-automation platform, deployed to **ECR/Lambda** across multiple **AWS** accounts.
- Integrate **LLMs** into production pipelines for extraction, classification, and natural-language generation, with guardrails, deterministic fallbacks, and cost/latency controls.
- Built a multi-tenant WhatsApp travel-booking agent (**Twilio** → **FastAPI** → **LangChain** tool-loop → **Duffel**) with **PostgreSQL row-level security** and a deterministic conversation-memory layer that recalls context without fabricating data.
- Diagnosed a **56% LLM** tool-call failure rate from **Braintrust** traces and eliminated it with a provider fallback (**Groq** → **Claude**) and date-grounding fixes, hardening the system to **274** automated tests with **mypy --strict** across **120+** files.
- Engineer for reliability and observability: **idempotent processing**, **exactly-once semantics**, distributed state coordination, structured logging, and correlation-ID tracing through **CloudWatch**.

Projects

AI SDR (Davis) [\[code\]](#) [\[live demo\]](#) **Next.js**, **React**, **TypeScript**, **FastAPI**, **LangGraph**, **Groq**, **Supabase**, **Pinecone**, **Redis**

- Built an evidence-grounded AI sales agent (**68** endpoints / **17** routers) that researches prospects and drafts sourced outreach; split scoring into an **LLM** signal-extraction stage and a deterministic **Python** scorer (**6** weighted dimensions summing to **100**) for byte-identical, unit-testable scores, with an approval-first send pipeline and **6**-rule policy guard preventing duplicate sends.
- Enforced multi-tenant isolation with **40 row-level-security** policies across **45 Postgres** tables (**0** unprotected), a **12**-intent reply classifier with **1.0** unsubscribe recall as a CI gate, and a **103**-case eval suite; deployed on **Vercel** and **Railway** with **Braintrust** tracing.

AdaptQuiz API [\[code\]](#) [\[live demo\]](#)

Python, **FastAPI**, **LangChain**, **FAISS**, **GPT-4o**, **Docker**, **Railway**

- Built and deployed a production **RAG** (Retrieval-Augmented Generation) API that turns PDF/TXT material into adaptive quizzes grounded in the source text at **\$0** embedding cost (**100%** local **all-MiniLM-L6-v2**, **384**-dim), with an **LLM-as-judge** grader (**GPT-4o**, **3**-criterion rubric, structured JSON) served over **6 REST** endpoints, containerized with **Docker** on **Railway** across **26** commits.

Focus-Flow [\[code\]](#)

Python, **FastAPI**, **SQLModel**, **SQLite**, **Next.js**, **TypeScript**

- Built a full-stack productivity app with a **FastAPI** + **SQLModel/SQLite** backend and a **Next.js/TypeScript** frontend, exposing **REST** endpoints for task decomposition, focus-session tracking, and usage statistics.

Technical Skills

Languages: Python, SQL, JavaScript / TypeScript

AI & LLM: LLM APIs (OpenAI GPT-4o, Groq, Anthropic Claude), RAG (Retrieval-Augmented Generation), LangChain, LangGraph, AI Agents, Prompt Engineering, Structured Outputs, Tool / Function Calling, LLM-as-Judge Evaluation, Embeddings (sentence-transformers), Semantic Search, FAISS, Pinecone

Backend: FastAPI, REST API Design, Pydantic, SQLAlchemy / SQLModel / Alembic, PostgreSQL, Supabase, DynamoDB, Redis, Event-Driven Microservices, Multi-Tenant Architecture, Row-Level Security, Next.js

Cloud & DevOps: AWS (Lambda, ECS, Step Functions, SQS/SNS, S3, DynamoDB, RDS, ECR, CloudWatch), Docker, Railway, Vercel, Git & GitHub, CI/CD

Tools & Foundations: pytest, Braintrust, OpenAPI / Swagger, NumPy, Pandas, ML Evaluation Metrics, Linear Algebra, Statistics

Education

Master of Computer Applications (MCA)

2022 – 2025

Indira Gandhi National Open University (IGNOU), New Delhi

Coursework: Software Engineering, Data Science, Cloud Computing, Artificial Intelligence & Machine Learning

B.Sc. (Hons) Physics with Mathematics

2017 – 2020

Jamia Millia Islamia, New Delhi

First Class Honours